

Research Notes on Convolutional neural network

Convolutional neural network

$$a^{(l)} = \sigma(W^{(l)}a^{(l-1)} + b^{(l)})$$

>> Section 1

One of the simplest methods to prevent overfitting of a network is to simply stop the training before overfitting has had a chance to occur. It comes with the disadvantage that the learning process is halted.

>> Section 2

simple cells, whose output is maximized by straight edges having particular orientations within their receptive field complex cells, which have larger receptive fields, whose output is insensitive to the exact position of the edges in the field. Hubel and Wiesel also proposed a cascading model of these two types of cells for use in pattern recognition tasks.

>> Section 3

ReLU layer ReLU is the abbreviation of rectified linear unit. It was proposed by Alston Householder in 1941, and used in CNN by Kunihiro Fukushima in 1969. ReLU applies the non-saturating activation function $f(x) = \max(0, x)$. It effectively removes negative values from an activation map by setting them to zero. It introduces nonlinearity to the decision function and in the overall network without affecting the receptive fields of the convolution layers. In 2011, Xavier Glorot, Antoine Bordes and Yoshua Bengio found that ReLU enables better training of deeper networks, compared to widely used activation functions prior to 2011. Other functions can also be used to increase nonlinearity, for example the saturating hyperbolic tangent $f(x) = \tanh(x)$, $f(x) = |\tanh(x)|$, and the sigmoid function $\sigma(x) = \frac{1}{1 + e^{-x}}$. ReLU is often preferred to other functions because it trains the neural network several times faster without a significant penalty to generalization accuracy.

>> Section 4

Animal behavior detection CNNs have been applied in ecological and behavioral research to automatically detect and quantify animal behavior from visual data, enabling identification of animals, tracking of individuals, estimation of pose, and classification of specific actions such as feeding, and social interactions. Combined with multi-object tracking and temporal modeling, these systems can extract behavioral sequences over extended recordings, reducing reliance on manual annotation and

increasing throughput for studies of individual variation, social networks, and collective dynamics.

>> Section 5

Convolution in time The term "convolution" first appears in neural networks in a paper by Toshiro Homma, Les Atlas, and Robert Marks II at the first Conference on Neural Information Processing Systems in 1987. Their paper replaced multiplication with convolution in time, inherently providing shift invariance, motivated by and connecting more directly to the signal-processing concept of a filter, and demonstrated it on a speech recognition task. They also pointed out that as a data-trainable system, convolution is essentially equivalent to correlation since reversal of the weights does not affect the final learned function ("For convenience, we denote $*$ as correlation instead of convolution. Note that convolving $a(t)$ with $b(t)$ is equivalent to correlating $a(-t)$ with $b(t)$."). Modern CNN implementations typically do correlation and call it convolution, for convenience, as they did here.

>> Section 6

Max norm constraints Another form of regularization is to enforce an absolute upper bound on the magnitude of the weight vector for every neuron and use projected gradient descent to enforce the constraint. In practice, this corresponds to performing the parameter update as normal, and then enforcing the constraint by clamping the weight vector $w \rightarrow \frac{w}{\|w\|_2} \min(\|w\|_2, c)$ of every neuron to satisfy $\|w\|_2 < c$. Typical values of c are order of 3–4. Some papers report improvements when using this form of regularization.

>> Section 7

Human interpretable explanations End-to-end training and prediction are common practice in computer vision. However, human interpretable explanations are required for critical systems such as self-driving cars. With recent advances in visual salience, spatial attention, and temporal attention, the most critical spatial regions/temporal instants could be visualized to justify the CNN predictions.

>> Section 8

DropConnect DropConnect is the generalization of dropout in which each connection, rather than each output unit, can be dropped with probability $1 - p$. Each unit thus receives input from a random subset of units in the previous layer. DropConnect is similar to dropout as it introduces dynamic sparsity within the model, but differs in that the sparsity is on the weights, rather than the output vectors of a layer. In other words, the fully connected layer with DropConnect becomes a sparsely connected layer in

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which the connections are chosen at random during the training stage.

>> Section 9

A deconvolutional neural network is essentially the reverse of a CNN. It consists of deconvolutional layers and unpooling layers. A deconvolutional layer is the transpose of a convolutional layer. Specifically, a convolutional layer can be written as a multiplication with a matrix, and a deconvolutional layer is multiplication with the transpose of that matrix. An unpooling layer expands the layer. The max-unpooling layer is the simplest, as it simply copies each entry multiple times. For example, a 2-by-2 max-unpooling layer is $\begin{bmatrix} x \\ \end{bmatrix} \mapsto \begin{bmatrix} x & x & x & x \end{bmatrix}$. Deconvolution layers are used in image generators. By default, it creates periodic checkerboard artifact, which can be fixed by upscale-then-convolve.

>> Section 10

Translation equivariance and aliasing It is commonly assumed that CNNs are invariant to shifts of the input. Convolution or pooling layers within a CNN that do not have a stride greater than one are indeed equivariant to translations of the input. However, layers with a stride greater than one ignore the Nyquist–Shannon sampling theorem and might lead to aliasing of the input signal While, in principle, CNNs are capable of implementing anti-aliasing filters, it has been observed that this does not happen in practice, and therefore yield models that are not equivariant to translations. Furthermore, if a CNN makes use of fully connected layers, translation equivariance does not imply translation invariance, as the fully connected layers are not invariant to shifts of the input. One solution for complete translation invariance is avoiding any down-sampling throughout the network and applying global average pooling at the last layer. Additionally, several other partial solutions have been proposed, such as anti-aliasing before downsampling operations, spatial transformer networks, data augmentation, subsampling combined with pooling, and capsule neural networks.